

Today's Content:-

- Number System Basics
- Binary to Decimal & Decimal to Binary
- Addition of Binary Numbers
- Bitwise Operations
- Properties
- Single Number
- Left Shift
- Right Shift

Number System

$$7 \ 3 \ 4 \rightarrow 700 + 30 + 4 \rightarrow 7 \times 10^2 + 3 \times 10^1 + 4 \times 10^0$$

$$6 \ 5 \ 9 \ 4 \rightarrow 6000 + 500 + 90 + 4 \rightarrow 6 \times 10^3 + 5 \times 10^2 + 9 \times 10^1 + 4 \times 10^0$$

$$\textcircled{2} \ 4 \ 5 \ 6 \rightarrow 2 \times 10^3 + 4 \times 10^2 + 5 \times 10^1 + 6 \times 10^0$$

$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow$
 $10^3 \ 10^2 \ 10^1 \ 10^0$ unit's place / 1's place

Decimal Number System $\rightarrow$ Base:- power of 10
 $\rightarrow$ digit:- [0-9]

1) Octal

Bases will be powers of 8

$$\begin{array}{cccc} 8^3 & 8^2 & 8^1 & 8^0 \\ \uparrow & \uparrow & \uparrow & \uparrow \\ (0 & 1 & 3 & 2)_8 \end{array}$$

$$\rightarrow 0 \times (8^3) + 1 \times (8^2) + 3 \times (8) + 2 \times (1)$$

$$\rightarrow 64 + 24 + 2 \rightarrow 90$$

Digits:- [0-7]

$$\rightarrow (10101)_8 \checkmark$$

$$\rightarrow (6784)_8 \times$$

$\rightarrow$ 8 is not possible in octal.

Quiz

$$(1 \ 2 \ 5)_8$$

$\begin{array}{ccc} 8^2 & & 8^0 \\ \swarrow & \downarrow & \searrow \\ & 8^1 & \end{array}$

$$\Rightarrow 8^2 \times 1 + 8 \times 2 + 1 \times 5 \Rightarrow 64 + 16 + 5 = 85$$

2) Ternary

$\rightarrow$ powers of 3

$\rightarrow$ digits: [0-2]

$$(0 \ 2 \ 1 \ 0 \ 1)_3 \rightarrow 3^4 \times 0 + 2 \times 3^3 + 1 \times 3^2 + 0 \times (3^1) + 1 \times (3^0)$$

$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow$
 $3^4 \ 3^3 \ 3^2 \ 3^1 \ 3^0$

$$\rightarrow 0 + 54 + 9 + 0 + 1 \Rightarrow 64$$

3) Binary $\rightarrow$ power:- 2
 $\rightarrow$ digits:- [0-1]

$$(10110)_2 \rightarrow 16 + 4 + 2 = 22$$

$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow$
 $2^4 \quad 2^3 \quad 2^2 \quad 2^1 \quad 2^0$

$$(011010)_2 \rightarrow 16 + 8 + 2 \Rightarrow 26$$

$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow$
 $2^5 \quad 2^4 \quad 2^3 \quad 2^2 \quad 2^1 \quad 2^0$

// Decimal to Binary

2		28	$\rightarrow$	0
2		14	$\rightarrow$	0
2		7	$\rightarrow$	1
2		3	$\rightarrow$	1
2		1	$\rightarrow$	1
2		0		

$\uparrow$

$$(11100)_2 = 16 + 8 + 4 = 28$$

2		19	$\rightarrow$	1
2		9	$\rightarrow$	1
2		4	$\rightarrow$	0
2		2	$\rightarrow$	0
2		1	$\rightarrow$	1
2		0		

$\uparrow$

$$(10011)_2 = 16 + 2 + 1 = 19$$

$$\begin{array}{r|l}
 2 & 25 \rightarrow 1 \\
 2 & 12 \rightarrow 0 \\
 2 & 6 \rightarrow 0 \\
 2 & 3 \rightarrow 1 \\
 2 & 1 \rightarrow 1 \\
 & 0
 \end{array}$$

$$(11001)_2$$

Decimal $\rightarrow$ Binary
Binary $\rightarrow$ Decimal.

Adding 2 decimal Numbers. $\rightarrow c = \text{sum}/10, d = \text{sum}\%10$

$$\begin{array}{r}
 \begin{array}{cccc}
 1 & 1 & 1 & \\
 3 & 4 & 5 & 9 \\
 0 & 2 & 8 & 4 & 7 \\
 \hline
 0 & 6 & 3 & 0 & 6
 \end{array}
 \end{array}$$

$$\begin{array}{r}
 \begin{array}{cccc}
 1 & 1 & 1 & \\
 2 & 4 & 5 & 8 \\
 3 & 7 & 6 & 4 \\
 \hline
 0 & 6 & 2 & 2 & 2
 \end{array}
 \end{array}$$

Add 2 Binary Numbers.

$$c = \text{sum}/2 \quad \text{and} \quad d = \text{sum}\%2$$

$$\begin{array}{r}
 \begin{array}{cccccc}
 0 & 0 & 1 & 1 & 0 & \\
 1 & 0 & 1 & 1 & 0 & \\
 0 & 0 & 1 & 1 & 1 & \\
 \hline
 0 & 1 & 1 & 1 & 0 & 1
 \end{array}
 \end{array}$$

Quiz

	0	0	1	1	
0		0	0		1
	0	1	0	0	1
	1	1	1	0	0

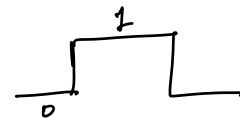
Q

$\frac{1}{2}$	$\frac{2}{2}$	$\frac{4}{2}$	$\frac{2}{2}$	$\frac{3}{2}$	$\frac{2}{2}$	$\frac{4}{2}$	
0	1	2	1	1	1	0	
			1	0	1	1	0 $\rightarrow 22$
			1	1	0	1	0 $\rightarrow 26$
			1	0	1	0	1 $\rightarrow 21$
$\frac{1}{2}$	$\frac{2}{2}$	$\frac{4}{2}$	$\frac{2}{2}$	$\frac{3}{2}$	$\frac{2}{2}$	$\frac{1}{2}$	
0	1	0	0	0	1	0	1 $\rightarrow 69$

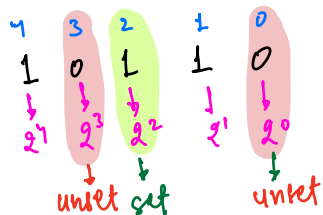
Ans $\Rightarrow (1000101)_2$

Naming Conventions:-

bit:
 → 1 : set/on
 → 0 : unset/off



e.g.



Bitwise Operators. (AND, OR, XOR, NOT, Left Shift, Right Shift)

$\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$
 $\&$ $|$ $\wedge$ $\sim$ $\ll$ $\gg$

Truth table.

a	b	$\&$	$ $	$\wedge$	$\sim a$ (Inverse operator)
0	0	0	0	0	1
0	1	0	1	0	1
1	0	0	1	0	0
1	1	1	1	1	0

$\downarrow$
 If both bits are set, then only o/p is set bit

$\downarrow$
 set if even one of them is set bit

$1 \leftrightarrow 0$

byte a = 28 :- $\begin{array}{ccccccc} 0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 \end{array}$
 byte b = 18 :- $\begin{array}{ccccccc} 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 \end{array}$ Decimal
 print(a & b) :- $\begin{array}{ccccccc} 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \end{array} = 16$
 print(a | b) :- $\begin{array}{ccccccc} 0 & 0 & 0 & 1 & 1 & 1 & 1 & 0 \end{array} = 30$
 print(a ^ b) :- $\begin{array}{ccccccc} 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 \end{array} = 14$
 print(~a) :- $\begin{array}{ccccccc} 1 & 1 & 1 & 0 & 0 & 0 & 1 & 1 \end{array}$

Properties

// byte a = 10

print(a & 1) →

Binary for a :-
 Binary for 1 :-

	7	6	5	4	3	2	1	0
a	0	1	0	1	0	0	0	0
1	0	0	0	0	0	0	1	0
	0	0	0	0	0	0	0	0

// byte a = 11

print(a & 1)

Binary for a :-
 Binary for 1 :-

	7	6	5	4	3	2	1	0
a	0	1	0	1	1	0	0	0
1	0	0	0	0	0	0	1	0
	0	0	0	0	0	0	1	0

= 1

if (a & 1 == 1) {

// a is odd number

// 0th bit is set

} v. imp.

$$a \& 0 = 0 \text{ (Always)}$$

$$a \& a = a$$

$$a | 0 = a$$

$$a | a = a$$

$$a \wedge 0 = a$$

$$a \wedge a = 0$$

1) $a \& b = b \& a$

2) $a | b = b | a$

3) $a \wedge b = b \wedge a$

Commutative Property

$$a \& b \& c = c \& b \& a$$

$$a \& (b \& c)$$

$$(b \& c) \& a$$

$$(c \& b) \& a$$

$$// a \& b \& c = a \& c \& b = b \& c \& a$$

$$// a | b | c = a | c | b = b | c | a$$

$$// a \wedge b \wedge c = a \wedge c \wedge b = b \wedge c \wedge a$$

Associative Property

$$Q \quad a^{\wedge} b^{\wedge} b^{\wedge} c^{\wedge} a = a^{\wedge} a^{\wedge} b^{\wedge} b^{\wedge} c = c$$

$$Q \quad c^{\wedge} d^{\wedge} a^{\wedge} c^{\wedge} a^{\wedge} g^{\wedge} d = g$$

Q Given N array elements, where every element repeats twice except 1.
Find the unique element.

↑
even times.

eg. arr[7] = 3 8 4 8 3 7 4 → ans = 7

eg. arr[5] = 2 2 7 9 7 → ans = 9

int uniqueElement (arr[], N) {

```

    ans = 0;
    for (i = 0; i < N; i++) {
        ans = ans ^ arr[i];
    }
    return ans;
}
```

ans = 0

$$0^{\wedge} 3 = 3$$

$$3^{\wedge} 8 = 11$$

$$11^{\wedge} 4 = 15$$

$$15^{\wedge} 8 = 7$$

$$7^{\wedge} 3 = 4$$

$$4^{\wedge} 7 = 3$$

$$3^{\wedge} 4 = 7$$